# Genuine Multipartite Entanglement Analysis

*Hubbard ring - Convex-roof framework*

## System parameters

| | |
|---|---|
| Model | **Hubbard 1D-PBC** |
| Mode | **uniform** |
| Sites (L) | **6** |
| Electrons (Ne) | **6** |
| Hopping (t) | **1.0** |
| Interaction (U) | **4.0** |
| n-body order | **3** |
| Eigenstate idx | **7 (1-indexed: 8)** |

## Eigenstate properties

| | |
|---|---|
| Energy E0 | **-1.76902482** |
| Energy E1 | **-1.76902482** |
| Gap (E1-E0) | **0.000000** |
| <S^2> | **0.000000** |
| S | **0.0000** |
| <Sz> | **0.000000** |
| Multiplicity | **1.0** |
| Spin label | **Singlet (S=0)** |

## Double occupancy

| | |
|---|---|
| <D> total | **0.351967** |
| <D> / site | **0.058661** |

## RDM and GME statistics for this run (3-body clusters)

Hilbert space dim : 2^3 = 8
Number of clusters : 0
Pure clusters : 0
Mixed clusters : 0
Clusters with GME (>0) : 0

Purity gamma = Tr(rho^2) : mean = — [min — , max —]
Von Neumann entropy : mean = — [min — , max —]
Rank of rho : mean = — [min — , max —]
C_GME (best estimate) : mean = — [min — , max —]

**NOTE** Analysis performed on eigenstate index 7 (state 8 in 1-indexed convention), not the ground state.

**Eigenstate index 7  (state 8 in 1-indexed convention)  -  NOT the ground state**

# Summary: All 3-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0  |  $E_0$=-1.769025

| Sites | Purity | C_GME | C_LB | C_cayl | C_expo | C_stiefel | C_UB | Status |
|---|---|---|---|---|---|---|---|---|

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Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map |  BEST
  C_UB = Upper bound (eigendecomp) | Status: GME  =has GME | NO GME=separable | MIXED=mixed state